Static Structural Analysis of an Axle Gearbox Housing

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1. Executive Summary

This report details the static structural analysis of a gearbox housing subjected to a 50 kN load. To optimize computational resources, a half-symmetry model was utilized. The primary objectives were to evaluate the structural integrity, identify and verify stress concentrations, and extract reaction forces at the mounting locations. The maximum equivalent stress was found to be approximately 135 MPa, **resulting in a minimum Factor of Safety of 1.93 against yield, indicating the structure safely carries the operational load**. Mesh refinement and path evaluations confirmed this as a legitimate stress concentration rather than a mesh singularity. Finally, internal moments and nodal forces were evaluated to validate the load path and provide recommendations for design optimization.

2. Model Setup & Boundary Conditions

2.1 Geometry and Contacts

The simulation utilizes a half-symmetry CAD model. To accurately represent the stiffness of the assembled components, all internal connections were established using bonded contacts. The material assigned to the housing is **Structural Steel with a yield strength of 260 MPa.**.

2.2 Boundary Conditions & Loads

The boundary conditions were established to mimic the physical constraints of the full assembly:

- **Symmetry:** A Named Selection was created for the symmetric face, and a Frictionless Support was applied to prevent translation normal to the cut plane.
- **Fixed Support:** Fixed supports were applied to the rear mounting holes. Due to the half-symmetry model, this corresponds to 3 full holes and 1 half-hole (representing the 7 holes on the physical model).
- **External Load:** A force of 50,000 N was applied at the rolling bearing faces in the positive Z-direction.
- **Analysis Settings:** Nodal forces were enabled in the solver settings to calculate reaction forces across structural cuts and at the individual bolt holes.

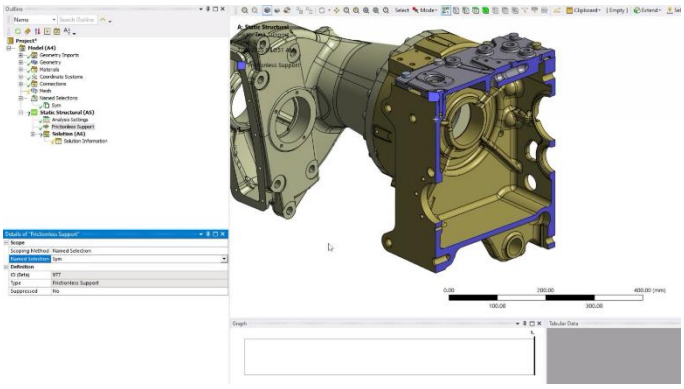


Figure 1: Application of half-symmetry frictionless support and 50 kN external load.

3. Meshing Strategy & Singularity Checks

3.1 Initial Mesh

An initial mesh was generated to check the global deformation and ensure loads and boundary conditions were defined correctly. This global mesh utilized a standard element size of 10 mm and a defeature size of 10 mm, **resulting in approximately 261974 nodes and 149188 elements.**.

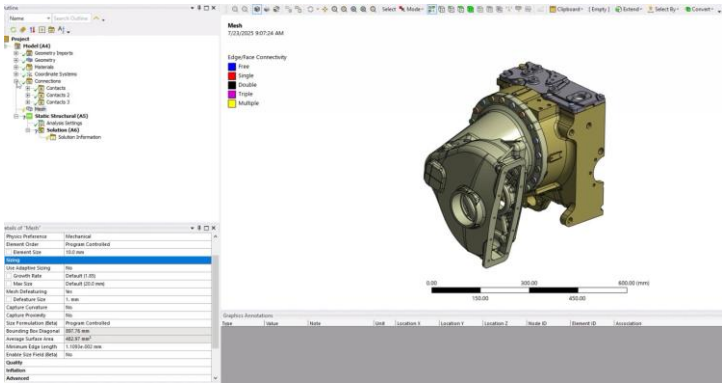


Figure 2: Initial global mesh with a 10 mm element size.

3.2 Localized Refinement & Singularity Evaluation

Initial results indicated high-stress peaks near the top bolt holes and a sharp edge, raising concerns about potential mesh singularities. To evaluate this, localized mesh refinements were applied. First, an element face sizing of 1 mm was applied to the face experiencing high stress. Second, for a more complex geometric location, a local coordinate system was created, and a Sphere of Influence was applied with a radius of 15 mm and an element size of 2 mm.

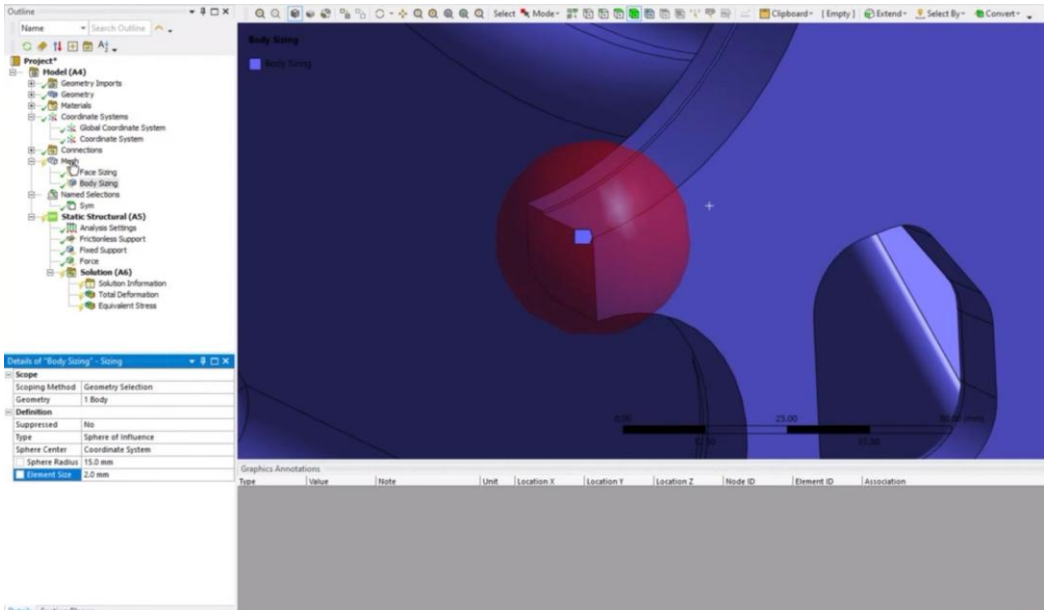


Figure 3: Application of a Sphere of Influence (15mm radius, 2mm element size) for targeted mesh refinement.

Following the second simulation with the refined mesh, the peak stress was re-evaluated. The results showed a uniform stress gradient peaking at roughly 135 MPa. Because the stress stabilized into a smooth color gradient rather than spiking infinitely at a single node, this confirmed the area is a legitimate stress concentration and not a numerical singularity.

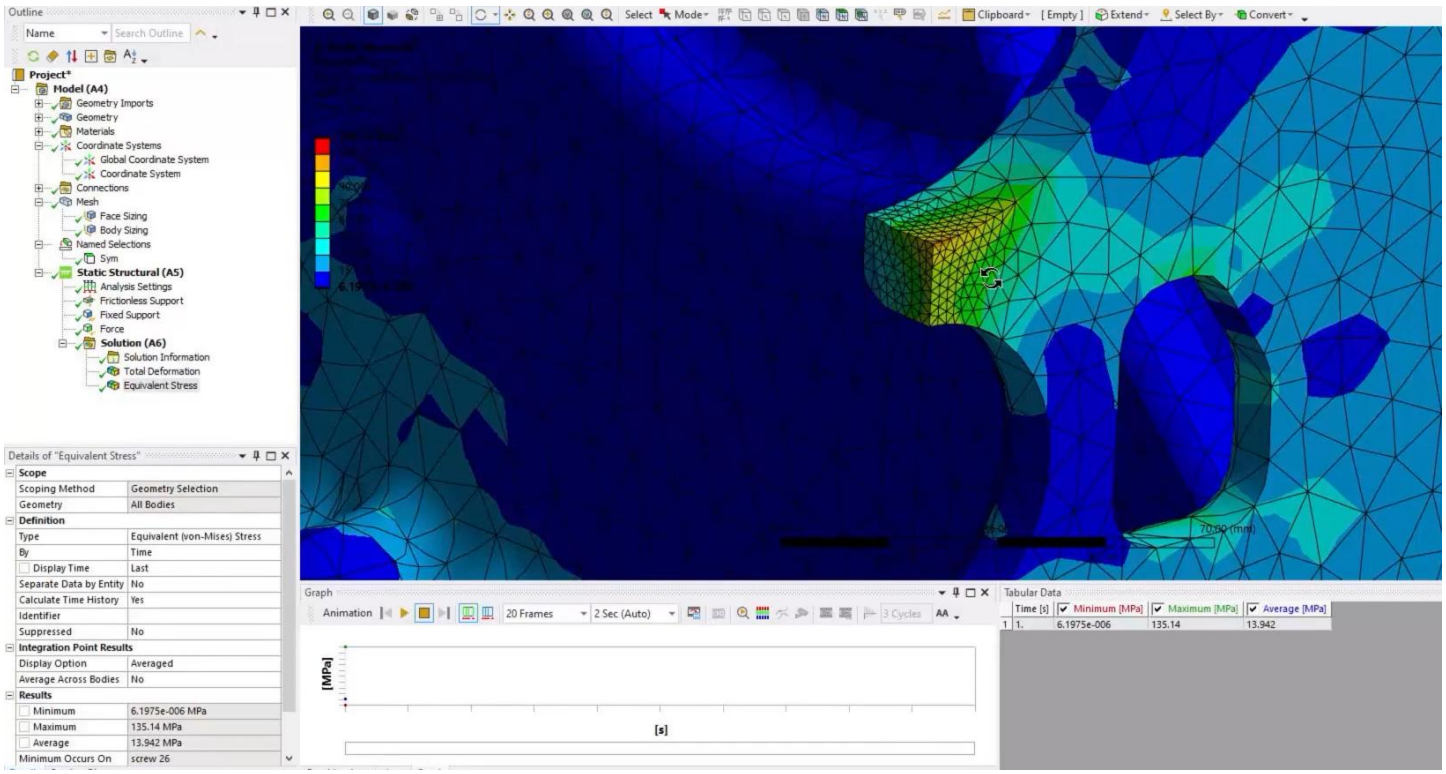


Figure 4: Equivalent (von-Mises) stress contour showing a uniform gradient peaking at 135 MPa, confirming a valid stress concentration.

4. Results & Discussion

4.1 Path Evaluation on Sharp Edges

To definitively rule out a mesh singularity at the sharp edges at the back of the housing, a scoping path was created directly along the edge geometry. The Equivalent (von-Mises) stress was evaluated across the length of this path (approximately 100 mm in length).

As shown in the plotted results, the stress profile demonstrates a gradual, uniform curve, rising from a baseline of approximately 9.8 MPa and peaking smoothly at 110.02 MPa. Because the curve is smooth and asymptotes naturally without exhibiting unphysical, jagged spikes, we can confidently conclude that this is a valid stress concentration and not a numerical singularity. However, because it is a sharp edge, adding a small fillet radius in future design iterations will ensure the stress is evaluated with even higher accuracy and will physically reduce the stress riser.

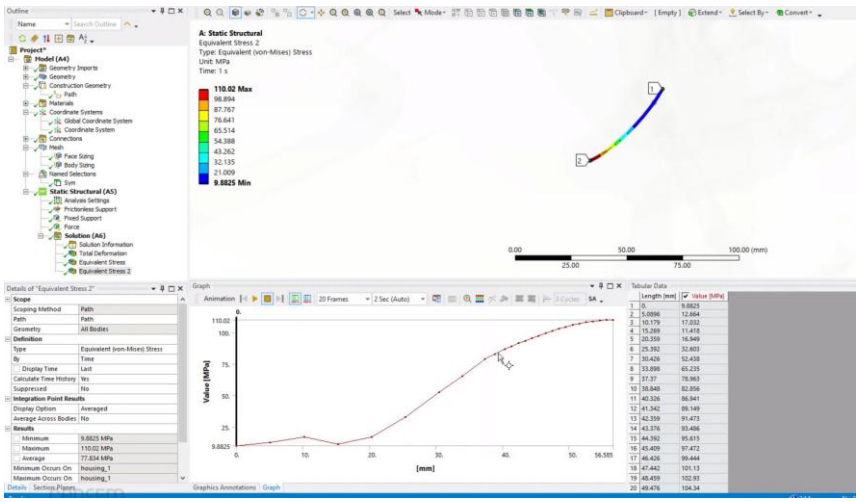


Figure 5: Stress evaluation along the defined edge path, demonstrating a smooth, uniform curve peaking at 110 MPa.

4.2 Internal Section Forces and Moments

To evaluate internal cutting forces, a new coordinate system and section plane were created. Nodal forces across this surface were evaluated as vectors. The results indicated that the X and Y direction forces were minimal (zero), properly reflecting the pure Z-direction applied load. Furthermore, a significant internal moment of approximately 6000 Nm acts along this structural plane.

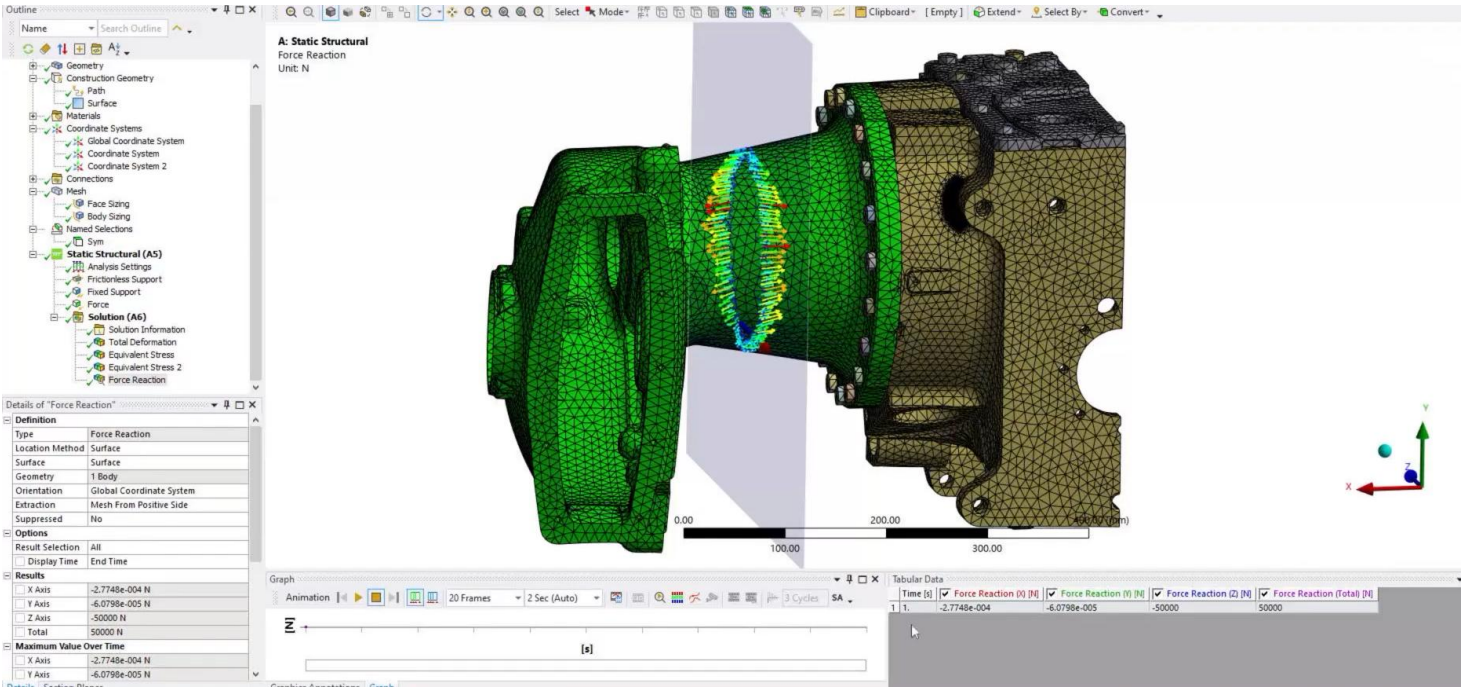


Figure 6: Evaluation of nodal forces and internal moments across the generated section plane.

4.3 Reaction Forces at Mounting Holes

To understand the load distribution and validate the structural load path, the force reaction at each individual mounting hole was extracted. Because the applied 50 kN load induces both a translation force and a significant bending moment (approx. 6000 Nm) across the housing, the fixed holes must react to both push and pull forces.

The forces absorbed by the holes on the half-symmetry model are summarized in the table below:

Table 1: Reaction Forces at Fixed Mounting Holes

Mounting Location	Reaction Force Absorbed	Notes
Top Hole	~ 40.9 kN	Absorbs the maximum reaction force.
Second Hole	17.0 kN	
Third Hole	7.5 kN	
Bottom Hole (Half)	12.5 kN	Due to symmetry, the full bottom hole absorbs 25.0 kN.

It is important to note that the sum of the individual bolt reaction forces (approx. 77.9 kN for the half-model) significantly exceeds the 50 kN applied shear load. Because the offset bearing load induces a massive 6000 Nm bending moment across the housing, the fixed bolts must counteract

this moment via a push-pull tension couple, rather than just resisting simple shear. The extraction of these nodal forces reveals a direct correlation between this load distribution and the structural stress. The uppermost hole absorbs the vast majority of the applied force (approx. 40.9 kN in tension) This correlates perfectly with the contour plots from Section 3.2, which identified the material immediately surrounding this top hole as the maximum stress hotspot (135 MPa).

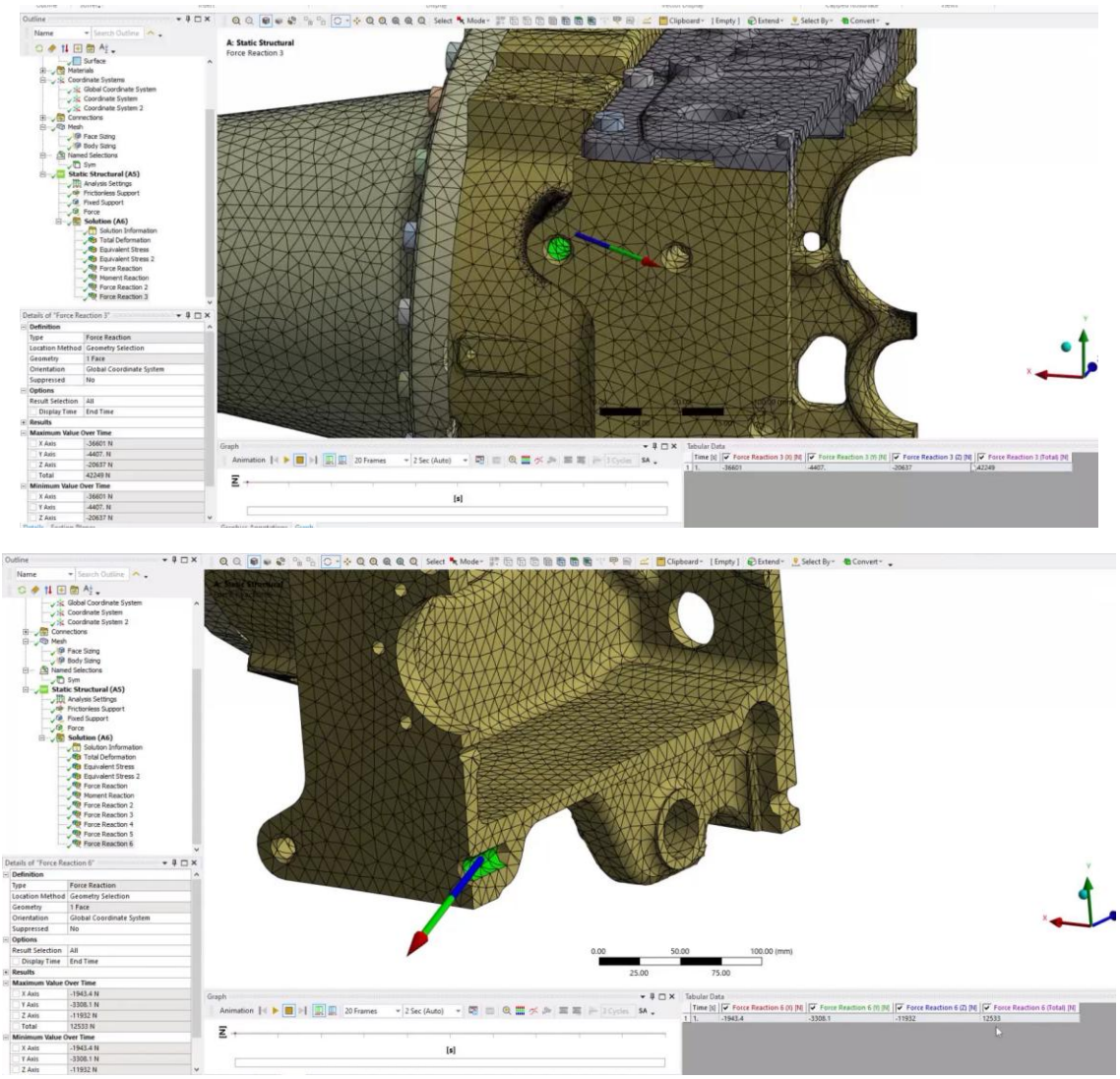


Figure 7: Extraction of reaction forces at individual mounting holes.

5. Conclusions & Recommendations

5.1 Conclusion

The static structural simulation successfully evaluated the load path and stress distribution of the gearbox housing. The maximum equivalent stress of 135 MPa is located near the upper mounting bolt, **providing a minimum Factor of Safety of 1.93.** Advanced scoping methods, including path mapping and targeted mesh refinement using a Sphere of Influence, confirmed this is a physical stress hotspot and not an artificial mesh singularity.

5.2 Recommendations for Design Improvement

Based on the simulation results, the following design improvements are recommended:

1. **Filleting:** Add a small radius to the sharp edges at the back of the housing to ensure stress is evaluated accurately and to alleviate localized stress risers.
2. **Geometry Modification:** If structurally possible within the larger assembly envelope, it is recommended to move the upper mounting hole slightly to the side. Because this upper hole absorbs the vast majority of the transferred force, relocating it would help distribute the load more evenly and significantly reduce the primary stress hotspot.

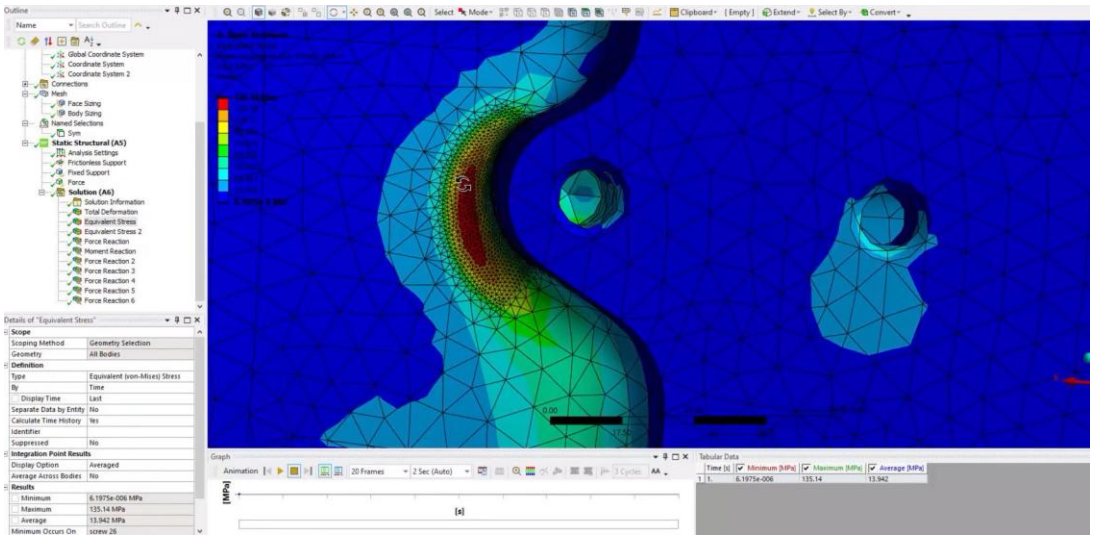


Figure 8: Primary stress hotspot. Moving this hole location is recommended to optimize load distribution.